أسئلة سنوات

Q1) Proportional Band of controller is Expressed as ? Percentage

Q2) Element That Translate The controller output into an action on The final Element? Actuator

Q3) The ___ compute error signal and use it to produce control Action? Controller

Q4) The objective of control system is? make Error Exactly zero

Q5) The controller called on/off controller Because The final Element has only one states ON OR off (True or Fales)? True

Q6) in The figure The Response is ? ~~Overdamped~~ underdamped



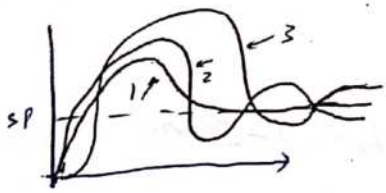
Q7) when you wait Temprature To reach setpoint That called? lag Time

Q8) Manipulated varible is The ___ that acted on by The controller? controlled varible

Q9) if Error Become more +ve as cv increase, The Action called? Direct Acting

Q10) if Error Become more +ve as cv decreased, The Action called? Reverse Acting

Q11) in The figure, what is Type of Response ?



1 - critically → على sp
2 - overdamped → يكون فوق sp
3 - underdamped → sp يكون تحت

Q12) one of the following is open loop system? Niddle valve

Q13) delay between change in input and initial Response of PV is? ~~Dead~~ lag Time

Q14) Allowable fluctuation in Error is ? Dead Band & Hysterisis

Q15) Two first order system with different lag Time conneted in cascade will produce ___ second order Response ? over damped

Q16) what is damping ratio for All Type of second order Response ?

overdamped $\frac{w}{z} > 1$, critically $\frac{w}{z} = 1$, Under $\frac{w}{z} < 1$

Undampal $\frac{w}{z} = 0$

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Q17) The unit of Time constant of system is the same of? Time

Q18) The Best control is larger duration with smaller Peak Error or vice versa (T&F)? True

Q19) The Difference Between SP & PV called? Error

Q20) when process Transfer function equal 1 that meaning? Dead Time is zero

Q20) if we increase PB what will happen? PB↑ then Kp↓ and offset ↑

Q21) What is process control? is the regulation of designed process within a specific target or set value called Setpoint → نقطة الضبط

Q22) The controller that we used in Air conditioning is? on-off controller

Q23) The main objective of Ziegler-Nichole closed loop method is To find value of? Proportional only gain that causes the control loop to oscillate indefinitely at a constant amplitude.

Q24) The Term reset controller is refer To? Integral controller

Q25) The percent change in Error that will cause 100% in controller output is? Proportional Band (PB)

Q26) Temperature sensor measure process Temperature and send result to controller To compare To setpoint, and controller Turn of Heater coil, This called? closed loop, Automatic control

Q27) Operator Turn of Heater coil when Temperature Transmitter output accertain Reading, this is called? closed loop, Manual control

Q28) controller Turnoff Heat coil ~~after~~ set interval, regardless of process Temperature, This is called? open loop, Automatic control.

Q29) Type of controller has largest steady state Error is? Proportional Controller

Q30) This symbol is for? Hydraulic line

~~LLLL~~

Q) Type of controller That repeat Action of Proportioned mode as long as Error Exist? PI (إذا كان) → (خط)

Q) Cascade control mean? more than one feed Back loop.

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Q33) The Tf of Ideal proportion plus reset controller? P+Reset is PI controller ⇒ Tf = $K_c(1+\frac{1}{Ts})$

Q34) We use ____ controller for flow loop when we need No offset? flow is fast & No offset. so we use PI controller

Q35) with formula $G_c(s) = 2(1+\frac{1}{4s})(1+0.25s)$ find K_P, K_D, K_I?

sol: $G_c(s) = H_c(s) = K_P + K_D s + \frac{K_I}{s}$

$= 2(1+\frac{1}{4s})(1+0.25s) = (2+\frac{1}{2s})(1+0.25s)$

$= 2 + 0.5s + \frac{1}{2s} + \frac{0.25}{2}$

$= 2.125 + 0.5s + \frac{0.5}{s}$ $K_P = 2.125 / K_D = 0.5 / K_I = 0.5$

Q36) The Settling Time Encounted with --- controller is shorter than the PID controller? PID

Q37) if we want to control Temperature of the Tank without offset, the most suitable controller can be used Type? PI + PID.

Q38) The ____ dynamic response is described as shortest settling Time, improve damping and zero state Error? PID

Q39) Proportional control provide change in CV Based on Time of History Error (True or Fales)? Fales

Q40) mode rate is controller Type of? derevative Because it's output is Based on rate of change of Measured variable

Q41) The process is (in figure)? Tank + Temperature

Q42) The use of 3 Position controller Tend To reduce cycling behavior of the control varible (True or fales)? True.

Q43) level control by using proportional controller is considerd as Reverse Acting (True or fales)? True

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أسئلة سنوات
مهمة جداً

Q44) The Percent change in Error That will cause 100% change in CV? Proportional Band

Q45) Type of Error That cared offset & Eventually offset is? Integral controller or PI

Q46) Type of controller That is Prone To leave offset? Proportional controller

Q47) Type of controller That Produce Erratic output in noisy application? Derivative

Q48) Type of controller That repeat Action of proportional as long as Error Exist?

Q49) the Pneumatic control valve symbol? 

Q50) mode rate is controller Type of? Derivative

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